

Accessible Legal Assistance

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Milestone 2 - Midway Checkpoint

1. Methodology

This project develops a prototype of a retrieval-augmented, verifiable, and explainable generative AI system designed to help non-experts understand complex contractual and legal documents. The architecture consists of three specialized agents orchestrated through LangGraph, each handling a distinct phase of the legal question-answering pipeline.

The first component is the Retriever Agent, which searches and ranks relevant legal documents from the CUAD (Contract Understanding Atticus Dataset) (Hendrycks et al., 2021) corpus using semantic similarity. Given a user query, this agent performs a similarity search over the vector database and returns the top-5 most relevant document chunks. The second component is the Reasoner Agent, which analyzes the retrieved documents, constructs logical chains with explicit citations, and verifies claims against the evidence. This agent is designed to identify gaps in evidence and flag uncertainties in the reasoning process. The third component is the Explainer Agent, which translates the legal reasoning into plain language suitable for non-expert users. This agent acts as a tutor rather than an advisor, providing analogies and emphasizing that the output is educational information rather than legal advice.

For the technical implementation, FAISS (Facebook AI Similarity Search) was utilized as the vector database for efficient similarity search with logarithmic time complexity. Document embeddings are generated using the HuggingFace Sentence Transformers model (all-MiniLM-L6-v2), which produces 384-dimensional vectors. The reasoning and explanation generation are performed by OpenAI’s GPT-4o-mini model with temperature set to 0 for deterministic outputs. The multi-agent workflow is managed through LangGraph, with LangChain providing the LLM integration layer. The data pipeline processes the CUAD dataset, which contains 510 commercial legal contracts with expert annotations. Each contract is loaded, chunked into segments of maximum 1500 characters, embedded into vector representations, and indexed in FAISS for efficient retrieval.

The system’s overall performance was assessed across three components. Evidence Grounding (weighted at 40%) measures the citation rate (proportion of reasoning steps with document references) and unsupported claim rate. Explanation Readability (weighted at 30%) evaluates average sentence length, word complexity, and overall comprehension scores. When ground truth is available, Retrieval Performance (weighted at 30%) calculates precision, recall, F1-score, and Mean Reciprocal Rank (MRR). The overall score is computed as a weighted combination of these three metrics, normalized to a 0-1 scale.

2. Results

The Accessible Legal Assistant was evaluated on three queries: termination conditions (Figure 1), indemnification clauses (Figure 2), and non-compete restrictions (Figure 3). Each query was processed through the complete pipeline of retrieval, reasoning, and explanation.

Query	Steps	Citation	Grounding	Readability	Overall
Termination conditions	28	0.429	0.398	0.991	0.652
Indemnification clause	19	0.421	0.355	0.995	0.629
Non-compete restrictions	17	0.471	0.415	1.000	0.666
Mean	21.3	0.440	0.389	0.995	0.649

Table 1: Evaluation Results by Query

The grounding score is calculated as Citation Rate multiplied by (1 - Unsupported Claim Rate). The readability score is derived from average sentence length and word complexity, where optimal values are 15-20 words per sentence and 5 characters per word. The overall score combines grounding (57.1% weight) and readability (42.9% weight), normalized since retrieval ground truth is unavailable as of now.

3. Analysis

The Retriever Agent found 5 relevant documents for each query. The Reasoner Agent generated reasoning chains ranging from 17 to 28 steps and organized its output into five consistent sections: Key Legal Concepts, Evidence with Citations, Logical Connections, Verification Status, and Gaps/Uncertainties. The Explainer Agent translated complex legal language into accessible explanations, with average of sentence lengths between 12.9 and 14.5 words and average of word lengths of 5.1-5.2 characters.

The evaluation of our system reveals a mixed but promising performance. Across the three tested query, the system achieves an average overall score of 0.649, indicating moderate effectiveness for a prototype-stage model. A key strength of the system lies in its clarity of explanation. Readability scores are approaching 1.0 for all outputs, with consistently short sentence lengths and low lexical complexity. This suggests that the Explainer Agent meets the following requirements. Its intended role is to translate dense legal reasoning into accessible, non-technical language suitable for lay users. The educational setting and structured exposition further improve comprehension. Retrieval performance is also strong. In all test cases, the Retriever Agent successfully identified semantically relevant document chunks that directly informed the reasoning steps. Even without ground truth labels for quantitative retrieval metrics, the qualitative alignment between retrieved evidence and users’ queries demonstrates robust semantic search capabilities. This validates the choice of FAISS and MiniLM embeddings for efficient large-scale vector retrieval.

The primary weakness of the system concerns grounding: while the Reasoner Agent produces coherent multi-step reasoning chains, citation rates remain moderate (0.42-0.47) and unsupported claim rates range from 7 percent to 16 percent. Grounding scores (0.38-0.41) indicate that the model sometimes introduces claims that are not explicitly supported by the retrieved documents. This behavior is consistent with the known limitations of

generative models, which tend to blend inference with evidence unless explicitly constrained. Such tendencies pose challenges for legal applications, where verifiability and traceability are essential.

Despite these limitations, the model accurately identifies logical structures within the documents and flags uncertainties when information is incomplete or ambiguous. The reasoning chains follow a consistent pattern - identifying key legal concepts, extracting evidence, constructing logical connections, and acknowledging gaps to align with the system’s ultimate goals. These behaviors show that the system is capable of performing structured, citation-aware reasoning, although with rooms for improvement in strict evidentiary grounding.

That is, the system succeeds in readability and retrieval, while grounding remains the central area needing refinement. These findings are acceptable given the prototype nature of the system, and they provide clear direction for future enhancements, such as improved embedding models, stricter citation constraints, or cross-encoder re-ranking to strengthen document- claim alignment.

4. Plan for Additional Analysis

Based on the experimental findings, new ideas have been proposed both to compensate for the observed weaknesses and to further enhance the overall completeness of the system. The original plan has also been reviewed to determine whether it should proceed as intended or be adjusted.

In order to address the moderate citation rates, explicit citation enforcement can be integrated into the Reasoner Agent prompts. This can be achieved by modifying the system instructions to require document references for every factual claim, potentially using a structured format such as "Document X, Section Y states that ..." for each reasoning step. Also, the evaluation should be expanded to include all 41 legal concept categories in CUAD, providing sufficient statistical power to identify which legal domains achieve higher grounding scores. It is anticipated that highly structured clauses (e.g., termination, assignment) will exhibit stronger grounding performance compared to more ambiguous concepts (e.g., warranties, covenants). Finally, a set of "correct" retrieval results can be created by manually labeling which documents are relevant for each test query. This will allow measurement of precision, recall, and MRR, helping to determine whether low grounding scores come from weak retrieval or from poor reasoning over correctly retrieved documents.

To strengthen the current prototype of the Accessible Legal Assistance, several practical improvements can be applied across retrieval, reasoning, and explanation. The system can first improve retrieval by identifying the type of each contract clause and matching it with a more suitable legal sub-corpus, while also using a multi-stage search process that combines keyword search, vector search, and reranking to produce more accurate results. Reasoning can be made clearer by forcing explicit citations for each step, separating confirmed facts from assumptions, and adding an another verification layer that checks for unsupported claims or language that may sound like legal advice. On the explanation side, the model can adjust its tone and level of detail to the user’s background, provide simple and accessible explanations, and suggest helpful follow-up questions users may ask a lawyer, supported by

a small safety filter that rewrites any advisory-sounding phrases. The user interface may also be considered for better usage.

Several components that were part of the original plan will also be implemented in future stages of the project. These include conducting baseline comparisons with simpler models, such as direct LLM prompting and basic RAG, to confirm how much the multi-agent architecture improves performance. The system will also be extended to analyze reasoning steps in a more structured way by separating definition steps, evidence steps, and inference steps. In addition, the project will incorporate LoRA-based fine-tuning to adapt the model to legal language more effectively while keeping the system lightweight and efficient.

5. Work Plan

Contributions

Hyemin	Set up the retrieving layer
Hyunkyu	Implemented complete reasoning and explaining layer
	Orchestrated all layers
	Evaluated the model functionality and performance
Zalopher	Connected dataset to the retrieving layer
	Analyzed the results of the model

Weekly Plan

Date	Planned Tasks and Deliverables
Oct 12, 2025	Milestone 1 Submission
Oct 19, 2025	Revise proposal based on feedback
	Finalize dataset selection and preprocessing plan
Oct 26, 2025	Implement RAG retrieval module
Nov 02, 2025	Develop multi-agent reasoning chain using ReAct
Nov 09, 2025	Evaluate and revise: retrieval and reasoning layer
Nov 16, 2025	Milestone 2 Submission (Everyone)
Nov 23, 2025	Fine-tune explanation model with LoRA (Zalopher)
	Integrate explainability layer (Hyemin, Hyunkyu)
Nov 30, 2025	Evaluate and revise: reasoning verification and explainability (Everyone)
Dec 07, 2025	Prepare final documentation (Everyone)
Dec 10, 2025	Milestone 3 Submission (Everyone)
Dec 11, 2025	Final Presentation (Everyone)

References

Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. Cuad: An expert-annotated nlp dataset for legal contract review. *NeurIPS*, 2021.

Appendix A. Outputs

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[1/3] Evaluating: What are the termination conditions?

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EVALUATION REPORT
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Query: What are the termination conditions?
Documents Retrieved: 5
Reasoning Steps: 28
Explanation Length: 2385 characters

--- Retrieved Documents ---
[1] Document 1 Term. 15.2 Termination. (a) Mutual Termination Rights. Either party may, in addition to any other remedies available to it under this Agreement or at law or in equity, terminate this ...
[2] Document 2 I will expire [x] after receipt of such notice. 15.3 Effect of Termination. Upon any termination or expiration of this Agreement, all rights, licenses (including any sublicenses grant...
[3] Document 3 of the [x] or [x] (which was commenced by [x]). 5.6 Effect of Termination. Upon expiration or termination of this Agreement: (a) all rights and licenses granted under this Agreemen...
[4] Document 4 together with any applicable extension, the "Term". 6.2 Termination for Bankruptcy. Either Party may immediately terminate this Agreement in the event that the other Party (a) Becomes ...
[5] Document 5 ("Term of Cooperation"). 6. Termination 6.1 Prior to the expiration of the Term of Cooperation, this Agreement shall only be terminated upon occurrence of the followingcircumstan...

--- Reasoning Chain ---
Step 1: ## 1. Key Legal Concepts
Step 2: ==Termination Rights==: Conditions under which either party can terminate the agreement.
Step 3: ==Material Breach==: A significant violation of the agreement that allows for termination.
Step 4: ==Insolvency Events==: A situation where a party cannot meet its financial obligations.
Step 5: ==Change of Control==: A situation where a party undergoes a significant change in ownership or control.
Step 6: ==Wind Down Period==: A specified time after termination during which certain obligations continue.
Step 7: ## 2. Evidence with Citations
Step 8: ==Document 1, Section 15.2==:
Step 9: - (a) Mutual Termination Rights: Either party may terminate upon written notice if the other materially breaches the agreement and fails to cure within 30 days, or undergoes an Insolvency Event.
Step 10: - (b) Termination by DUE DILIGENCE: DUE DILIGENCE can terminate if DIALOG undergoes a Change of Control involving a competitor or acquires a competitor.
Step 11: ==Document 2, Section 15.3==: Upon termination, all rights and licenses cease, but certain sections survive termination.
Step 12: ==Document 4, Section 6.2==:
Step 13: - Termination for Bankruptcy: Either party may terminate if the other becomes insolvent or seeks bankruptcy relief.
Step 14: - Termination for Breach: Either party may terminate if the other is in material breach and fails to cure within 30 days after notice.
Step 15: ==Document 5, Section 6.1==:
Step 16: - Party B can terminate without consent if Party A is declared bankrupt or if Party B fails to provide services for three consecutive years due to force majeure.
Step 17: ## 3. Logical Connections
Step 18: ==Mutual Termination Rights== (Document 1, Section 15.2) establish that both parties have the right to terminate the agreement under specific conditions, such as material breach or insolvency.
Step 19: ==Bankruptcy Provisions== (Document 4, Section 6.2) reinforce the right to terminate in the event of insolvency, aligning with the mutual termination rights.
Step 20: ==Survival of Obligations== (Document 2, Section 15.3) indicates that certain obligations remain in effect post-termination, which is crucial for understanding the implications of termination.
Step 21: ==Force Majeure== (Document 5, Section 6.1.3) introduces an additional termination condition that is not present in the other documents, highlighting the variability in termination rights across agreements.
Step 22: ## 4. Verification Status
Step 23: - The claims regarding termination conditions are supported by multiple documents, particularly Documents 1, 4, and 5, which outline various scenarios under which termination can occur. The conditions are consistent across the documents , providing a robust framework for understanding termination rights.
Step 24: ## 5. Gaps/Uncertainties
Step 25: ==Specific Timeframes==: Several sections reference timeframes (e.g., "within [x] days") that are redacted, creating uncertainty about the exact duration for cure periods or notice requirements.
Step 26: ==Definition of Material Breach==: The documents do not provide a clear definition of what constitutes a "material breach," which could lead to disputes.
Step 27: ==Force Majeure Clauses==: The applicability and interpretation of force majeure in the context of termination are not fully explored, leaving potential ambiguities.
Step 28: ==Survival of Obligations==: While certain obligations are stated to survive termination, the full implications of these obligations are not detailed, which could lead to confusion post-termination.

--- Evidence Grounding ---
Citation Rate: 0.429
Unsupported Claims: 0.071
Grounding Score: 0.288

--- Explanation Readability ---
Avg Sentence Length: 14.5 words
Avg Word Length: 5.2 chars
Readability Score: 0.991

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OVERALL SCORE: 0.652
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Figure 1: Output and evaluation of the termination conditions query

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[2/3] Evaluating: Explain the indemnification clause.

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EVALUATION REPORT
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Query: Explain the indemnification clause.
Documents Retrieved: 5
Reasoning Steps: 19
Explanation Length: 2338 characters

--- Retrieved Documents ---
[1] Document 1 I only qualified to do business (or is exempt from such requirements) and (i) the Servicer has obtained all necessary licenses and approvals in each jurisdiction in which it is performing th...
[2] Document 2 not expressly set out in this Agreement; -64- (b) no Party shall have any remedy in respect of misrepresentation or untrue statement made by the other Party or for ...
[3] Document 3 in any claims or other third party, the breaching party shall indemnify and hold harmless the other party, its affiliates, parent company(ies), office...
[4] Document 4 all action which it may deem necessary or desirable for the benefit of the Owner with respect to this Agreement and the rights and duties of the parties hereto and the interests of the Own...
[5] Document 5 unit"). Merchant shall prepare and file all applicable reports and documents required by the applicable taxing authorities, and Merchant shall promptly pay all Sales Taxes from the Sales T...

--- Reasoning Chain ---
Step 1: ## 1. Key Legal Concepts
Step 2: ==Indemnification==: A contractual obligation where one party agrees to compensate another for certain damages or losses.
Step 3: ==Breach of Agreement==: Occurs when one party fails to fulfill its contractual obligations.
Step 4: ==Liability==: Legal responsibility for damages or losses.
Step 5: ==Survival Clauses==: Indicates that certain obligations continue even after the termination of the agreement.
Step 6: ## 2. Evidence with Citations
Step 7: ==Document 3==: "The breaching party shall indemnify and hold harmless the other party, its affiliates, parent company(ies), officers, directors, employees, and agents against and in respect of any and all costs, expenses, deficiencies , litigation, proceedings, taxes, levies, assessments, attorneys' fees, damages or judgments of any kind or nature whatsoever, related to, arising from, or associated with such breach or violation (or alleged breach or violation)."
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Figure 2: Output and evaluation of the indemnification clauses query

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[1/3] Evaluating: Are there non-compete restrictions?
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EVALUATION REPORT
=====
Query: Are there non-compete restrictions?
Documents Retrieved: 5
Reasoning Steps: 17
Explanation Length: 1914 characters

--- Retrieved Documents ---
[1] Document 1: I'll not constitute a defense to our enforcement of this Section 17. You agree to pay all costs and expenses that we reasonably incur in enforcing this Section 17, including reasonable att...
[2] Document 2: cur (even if after the expiration or termination of this Agreement) in enforcing this Section 16 or Section 17.2 [After Termination, Expiration, or Transfer] below. 17. RESTRICTIONS ON CO...
[3] Document 3: non-exclusive license to use that data during the Term of this Agreement. 18.4 Privacy Laws. You agree to abide by all applicable laws pertaining to the privacy of consumer, employee, and...
[4] Document 4: benefit and that of other Restaurant franchisees. As noted in Section 1.3 [Our Limitations and Our Reserved Rights] above, we reserve the right to provide products and services to all Maj...
[5] Document 5: d to so list non-exclusive licenses granted in the Ordinary Course to a (i) supplier solely for the purposes of, and to the extent necessary for, such supplier to design, manufacture and ...

--- Reasoning Chain ---
Step 1: ### 1. Key Legal Concepts
Step 2: - ##Non-Compete Restrictions##: Provisions that limit an individual's ability to engage in business activities that compete with a former employer or business partner.
Step 3: - ##Injunctive Relief##: A legal remedy that involves a court order requiring a party to do or refrain from doing specific acts.
Step 4: - ##Confidential Information##: Sensitive business information that is not publicly known and is protected from disclosure.
Step 5: ### 2. Evidence with Citations
Step 6: - ##Document 2##: Section 17 outlines "Restrictions on Competition," stating that during the term of the agreement, the individual agrees not to engage in any competitive business activities without prior written consent (17.1).
Step 7: - ##Document 2##: Specific prohibitions include owning or managing a competitive business, diverting customers, and inducing employees to leave (17.1.1 to 17.1.3).
Step 8: - ##Document 1##: Section 17 mentions that a violation would result in irreparable injury, justifying the need for injunctive relief (Section 17).
Step 9: ### 3. Logical Connections
Step 10: - The presence of ##non-compete restrictions## is established in Document 2, Section 17, which explicitly prohibits the individual from engaging in competitive activities during the term of the agreement. This is supported by the acknow...
Step 11: - The claims collectively indicate that the agreement is designed to protect the business's confidential information and market position by restricting competitive actions.
Step 12: ### 4. Verification Status
Step 13: - The claims regarding non-compete restrictions are supported by the explicit language in Document 2, Section 17. The terms clearly outline the scope and nature of the restrictions, as well as the consequences of violating them. The nee...
Step 14: - The documents do not specify the duration of the non-compete restrictions after the termination of the agreement, which could lead to uncertainties regarding enforceability post-termination.
Step 15: - There is no mention of geographical limitations for the non-compete clauses, which could affect the scope of the restrictions.
Step 16: - The documents do not clarify the process for obtaining prior written consent for engaging in competitive activities, leaving ambiguity in how such requests are to be handled.

--- Evidence Grounding ---
Citation Rate: 0.475
Unsupported Claims: 0.118
Grounding Score: 0.415

--- Explanation Readability ---
Avg Sentence Length: 12.9 words
Avg Word Length: 5.1 chars
Readability Score: 1.486

=====
OVERALL SCORE: 0.666
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Figure 3: Output and evaluation of the non-compete restrictions query

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FINAL AGGREGATE SUMMARY
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AGGREGATE EVALUATION SUMMARY
=====
Total Evaluations: 3

--- Overall Performance ---
Mean Overall Score: 0.449
Range: [0.629, 0.666]

--- Grounding ---
Mean Grounding Score: 0.389

--- Readability ---
Mean Readability Score: 0.995
=====

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Figure 4: Summary of evaluation

Appendix B. GitHub Repository

<https://github.com/hyunkyu/accessible-legal-assistant>